

Roll Number: _____

Indian Institute of Technology Madras
Department of Management Studies

MS6015: Multivariate Statistical Methods for Business
Sample Endterm Examination

1. This quiz has a total 19 questions worth 40 marks.
2. The duration of the quiz is 120 minutes.
3. Answers to the questions should be provided in the space provided only.
4. For questions, that require a numerical answer: *Give explanation (application of formula, if any, used) for the same. Marks will not be awarded for answers without explanation (or stating only formula used), even if correct.*
5. If you feel the question is incorrect, please mention the same in the space provided- give your reasoning.
6. Read all questions carefully.
7. *All the best.*

Section 1. Fill in blanks (1 mark each): Total 6 Marks

1. Generalised variance is _____ of covariance matrix.
2. A strong linear dependence between some of the covariates in a multiple regression model is called _____.
3. In a logistic regression model, the statistical significance of a predictor variable is tested using _____.
4. The multivariate generalization of chi-square distribution is _____.
5. Simple correlations between the variables and the factors is called _____.
6. _____ the Jaccard Similarity coefficient (JSC), higher the similarity between two observations being compared. The value of JSC lies between _____.

Section 2. Fill in blanks (2 mark each)-Justify your answer: Total 4 Marks

7. Consider the covariance matrix given by

$$\Sigma = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Assume that this covariance structure can be explained by a factor analysis model $X - \mu = LF + \epsilon$ with loadings $L = (2, 1, 0)'$. The specific variances are $\psi_1 = ______, \psi_2 = ______, \psi_3 = ______$

8. For a classification problem with two classes, the proportion of positives at a node is 20%. The value of the Gini index at this node is ____
Explanation: $GI(t) = \sum_{i=1}^K \sum_{j=1, j \neq i}^K P(C_i|t)P(C_j|t) = 1 - \sum_{i=1}^k [P(C_i|t)]^2 = 1 - 0.04 - 0.64 = 0.32$.

Section 3. Multiple Choice Questions (2 mark each)-Justify your answer:
Total 16 Marks

9. The eigen values in a factor analysis add up to 120. Two factors explain 20% of the variance. What is the total of the eigen values for other factors?

- (a) 96
- (b) 100
- (c) 117
- (d) 97

Explanation:

10. If $X \sim N_p(\mu, \Sigma)$, then any linear combination $a'X = a_1X_1 + a_2X_2 \dots + a_pX_p$ is distributed as

- (a) $a'X \sim N_p(\mu, \Sigma)$
- (b) $a'X \sim N_p(\mu, n\Sigma)$
- (c) $a'X \sim N_p(\mu, \frac{\Sigma}{n})$
- (d) None of the above

11. The human resources manager of a large corporation has analyzed data for 48 randomly selected employees, looking at health-care claims (\$HC) filed by the employees over the past year. She observed a difference in the average claims amounts between male and female employees. She suspects that a confounding variable may be the age (AGE) of the employee: on average, the males at the corporation are older than the female employees, and a scatterplot of \$HC vs. AGE shows a positive linear association. The manager is trying to decide if the difference in the average claims amounts between males and female employees may be explained to some degree by the age of the employees. An analysis of covariance is performed, using a dummy variable GENDER, coded as 1 for male and 0 for female. The other explanatory variable is AGE, and the interaction is GENDER $\times$ AGE. The conditions for multiple regression have been met. The results are provided:

	s_e	376.25		
	n	48		
ANOVA				
Source	SS	df	MS	F
Regression	87,996,469.6758	3	29,332,156.5586	208.31
Residual	6,195,738.3242	44	140,812.2346	
Total	94,192,208.0000	47		
Regression				
Variables	coefficients	std. error	t	p
Intercept	850.6213	277.2837	3.068	.0037
AGE	69.4012	6.9541	9.980	7.11E-13
GENDER	150.8540	400.1913	0.377	.7080
GENDER $\times$ AGE	22.1079	9.1182	2.425	.0195

Following the principle of marginality, determine the estimated average health-care claims for a male employee who is 50 years old. (Round your answer to one decimal place.)

- (a) \$2106.9
- (b) \$5576.9
- (c) \$5426.1
- (d) \$8415.4

12. Is there a statistically significant difference between the slopes on AGE for men and women?
- (a) No, GENDER has a p-value = 0.7080 > 0.05, so we conclude the slope for men and women are equal.
 - (b) Yes, AGE has a p-value = 7.11E-13 < 0.05, so we conclude the slope for men and women are significantly different from one another.
 - (c) No, The confidence interval for GENDER includes 0 and therefore there is no significant difference in the genders.
 - (d) Yes, AGE $\times$ GENDER has a p-value = .0195 < 0.05, so we conclude the slope for men is predicted to be greater than the slope for women.
13. The operations manager of a bank is interested in any association that may exist between the time required to process a loan and the dollar amount of the loan. Since the bank makes a wide variety of loans, from autos to home repair to small business, there is a large amount of variation in the time required to process the loans. Using the explanatory variable *AMT* (in thousands of dollars) for the dollar amount of the loan, and the response variable *TIME* (in days per loan) for the time required to process the loan, a simple regression model was attempted. The scatterplot indicated a curved pattern, so a reciprocal transformation was performed on the response variable so that the model becomes:

$$\frac{1}{Time} = b_0 + b_1 \times AMT + \epsilon$$

. (Note the units on the response are loans per day.) The software output provided the following results:

$$Estimated \frac{1}{Time} = 241 - 0.001 \times AMT; R^2 = 0.507, \rho = -0.712, s_e = 0.092, n = 29$$

Using this model, what is the estimated average time required for processing loans of \$200,000? (Round your answer to 1 decimal place.)

- (a) 24.4 days
- (b) 8.2 days
- (c) 48 days
- (d) This value cannot be estimated without the value for ϵ being given.

A binary logistic regression model is developed for probability of default in payment as dependent variable and the credit score as the predictor variable. The corresponding logit function is given by:

$$\ln \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = 15 - 0.025 \times \text{credit score}$$

$Y = 1$ indicates default. Answer the three questions below using the above equation

14. The value of credit score for which the probability of default in payment and probability of no default are equally likely is
 - (a) 575
 - (b) 600
 - (c) 450
 - (d) 500

15. A bank declines loan application when the probability of default is 0.2 (or 20%) or higher. The minimum credit score required to secure a loan from this bank is approximately
 - (a) 675
 - (b) 656
 - (c) 625
 - (d) 666

16. CHAID tree uses the following criteria for splitting:
- (a) Chi-square Goodness of Fit Test for continuous predictor and F -test for discrete predictor
 - (b) Chi-square Test of Independence for all types of predictor variables
 - (c) Chi-square Test of Independence for discrete dependent variable and F -test for continuous dependent variable.
 - (d) F -test for all type of predictor variables

Section 4. Multiple Choice Questions -Justify your answer:
Total 4 Marks

17. After three iterations of Hierarchical Agglomerative Clustering using Euclidean distance between points, we get the 3 clusters: $C1 = \{2, 4\}$, $C2 = \{7, 8\}$ and $C3 = \{12, 14\}$. What is the distance between clusters $C1$ and $C2$ using Single Linkage and Complete Linkage?
- (a) 2,2
 - (b) 3,4
 - (c) 3,6
 - (d) 5,6

Section 5. Short Answer (5 marks each)- Total 10 Marks

18. Twenty sampling units are randomized into 4 treatment groups, with 5 units in each group. Two measurements are then taken from each unit. The sample group means and variance-covariance matrices appeared to be:

$$\text{Group 1: } \bar{X}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}; S_1 = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

$$\text{Group 2: } \bar{X}_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}; S_2 = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

$$\text{Group 3: } \bar{X}_3 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}; S_3 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\text{Group 4: } \bar{X}_4 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}; S_4 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Conduct the MANOVA test at the 5% level of significance to see if there are any differences among four multivariate population group means. In particular,

- a State the null and alternate hypothesis. [1 mark] .

- b Using the information in Part a, construct the one-way MANOVA table.

$$\begin{aligned} \text{Hint: } \sum_{l=1}^k \sum_{j=1}^{n_l} (x_{lj} - \bar{x})(x_{lj} - \bar{x})' &= \sum_{l=1}^k n_l (\bar{x}_l - \bar{x})(\bar{x}_l - \bar{x})' + \sum_{l=1}^k \sum_{j=1}^{n_l} (x_{lj} - \bar{x}_l)(x_{lj} - \bar{x}_l)' \\ \text{Total(corrected)SSCP} &= \text{Treatment} + \text{Residual} \\ &= \underline{\text{Between Groups}} + \underline{\text{Within Groups}} \end{aligned}$$

W is proportional to a pooled estimated of the common Σ

[2 marks]

c Evaluate Wilks' lambda.

[2 marks]

19. Technical and artistic scores of figure skaters are correlated, and therefore, they can probably be represented by one principal component. A sample of skaters yields covariance matrix

$$S = \begin{bmatrix} 0.5 & 0.2 \\ 0.2 & 0.8 \end{bmatrix}$$

The eigen values of S are $\lambda_1 = 0.9, \lambda_2 = 0.1$. The normalized eigen vector corresponding to λ_1 is $e_1 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$

- a. What is the percentage of variance explained by the first principal component. [1 mark]

- b. Two competing figure skaters receive the scores $X_1 = (5.5, 5.9)$ and $X_2 = (5.7, 5.7)$. What would be their scores according to the first principal component? Is there a clear winner in this case? Explain. [4 marks]

Rough

Answer Key for Exam A

Section 1. Fill in blanks (1 mark each): Total 6 Marks

1. Generalised variance is determinant of covariance matrix.
2. A strong linear dependence between some of the covariates in a multiple regression model is called multicollinearity.
3. In a logistic regression model, the statistical significance of a predictor variable is tested using Wald's test.
4. The multivariate generalization of chi-square distribution is Wishart distribution.
5. Simple correlations between the variables and the factors is called Factor Loadings.
6. Higher the Jaccard Similarity coefficient (JSC), higher the similarity between two observations being compared. The value of JSC lies between 0 and 1.

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